

Smart Society Backend Tech Stack

Language

- Python 3.11+
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Backend Framework

- Flask

Purpose:

- REST API development
 - Application framework
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REST API Framework

- Flask-RESTX

Purpose:

- REST API development
 - Swagger UI
 - OpenAPI Documentation
 - Request Models
-

Database

- PostgreSQL

Development

- Neon PostgreSQL (Cloud)

Deployment

- Neon PostgreSQL
-

ORM

- Flask-SQLAlchemy
- SQLAlchemy

Purpose

- Database Models
 - Relationships
 - Query Builder
 - ORM
-

Database Migration

- Flask-Migrate

Purpose

- Database Version Control
- Schema Migration

Commands

flask db init

flask db migrate

flask db upgrade

Authentication

- Flask-JWT-Extended

Purpose

- JWT Authentication
- Access Tokens
- Protected APIs
- Role Based Access Control

Password Hashing

- Werkzeug Security

Functions

`generate_password_hash()`

`check_password_hash()`

Validation

- Marshmallow

Purpose

- Request Validation
 - Response Serialization
 - Email Validation
 - Input Sanitization
-

Cross Origin Requests

- Flask-CORS

Purpose

Frontend (React)

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Backend (Flask)

Environment Variables

- `python-dotenv`

Purpose

DATABASE_URL

SECRET_KEY

JWT_SECRET_KEY

UPLOAD_FOLDER

File Upload

- Werkzeug

Purpose

- Complaint Images
- Completion Photos
- Attachments

Storage

uploads/

API Documentation

- Swagger UI

Generated using

- Flask-RESTX

Full Backend Architecture

React Frontend



Axios



Flask REST APIs



Routes



Controllers



Services



Schemas (Validation)



SQLAlchemy Models



PostgreSQL (Neon)

Models

- User
 - Category
 - Complaint
 - ComplaintUpdate
 - Attachment
 - Notification
 - Feedback
-

Planned API Modules

Authentication

- Register
 - Login
 - Logout
 - Profile
-

User

- Current User
-

Resident

- Raise Complaint
 - Complaint History
 - Complaint Timeline
 - Feedback
-

Admin

- Dashboard
- Complaint Management

- Assign Staff
 - Reports
 - Analytics
 - Category Management
-

Staff

- Assigned Complaints
 - Update Status
 - Work History
 - Notification Preferences
-

Notifications

- Notification Center
 - Read Notification
 - Mark All Read
 - Notification Settings
-

Security

- JWT Authentication
 - Password Hashing
 - Role-Based Authorization
 - Environment Variables
 - Input Validation
 - CORS
 - Secure Password Storage
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Database Tables

- users
- categories
- complaints

- complaint_updates
 - attachments
 - notifications
 - feedbacks
-

Development Tools

- VS Code
 - Git
 - GitHub
 - Postman
 - pgAdmin (optional)
 - Neon PostgreSQL
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Deployment (Planned)

Backend

- Render

Database

- Neon PostgreSQL

API Documentation

- Swagger UI